

PROJECT OVERVIEW AND DATA PRIVACY ASSURANCE

Project Title: Fitonexa: AI-Assisted Physiotherapy Exercise Assessment System

Institution: Vivekanand Education Society's Institute of Technology (VESIT), Mumbai

Department: Automation and Robotics Engineering

Faculty Supervisor: Dr. Sangeetha Prasannaram, Head of Department

1. Project Description

Fitonexa is an academic research project being developed by students of the Department of Automation and Robotics Engineering at Vivekanand Education Society's Institute of Technology (VESIT), Mumbai.

The objective of this project is to develop an edge-based intelligent system that assists physiotherapy patients and practitioners by analyzing exercise performance in real time. The system captures video frames of patients performing prescribed physiotherapy exercises and utilizes Computer Vision and Machine Learning techniques to:

- Identify body joints and skeletal landmarks.
- Calculate relevant joint angles and movement parameters.
- Classify physiotherapy exercises being performed.
- Evaluate the correctness of exercise execution.
- Provide corrective feedback regarding posture, alignment, and movement patterns.

The ultimate goal of the project is to improve exercise adherence, reduce the risk of incorrect movements, and support physiotherapy rehabilitation through technology-assisted feedback.

To train, validate, and improve the accuracy of the system, real-world exercise data collected under the supervision of qualified physiotherapists is required.

2. Data Collection Requirements

Subject to approval from the clinic and voluntary patient participation, the project team may collect:

- Video recordings of patients performing physiotherapy exercises.
- Body pose information and skeletal landmarks extracted from video frames.
- Joint angle measurements and movement-related parameters.
- Exercise labels and observations provided by physiotherapists.

The focus of data collection is solely on movement analysis and exercise performance.

3. Data Privacy and Confidentiality Commitment

The project team is committed to maintaining strict confidentiality and protecting patient privacy throughout the project.

The following measures will be implemented:

a) No Personal Information Collection

- Patient names, phone numbers, addresses, medical record details, and other personally identifiable information will not be collected or stored.
- Each participant will be assigned an anonymous identification code for research purposes.

b) Face Anonymization

- Patient faces will be blurred, masked, cropped, or otherwise anonymized before any data is used for analysis, storage, presentations, demonstrations, or publications.
- The final dataset used for machine learning model development will not contain identifiable facial information.

c) Secure Data Storage

- All collected data will be stored securely on password-protected systems accessible only to authorized project members and faculty supervisors.
- Appropriate safeguards will be implemented to prevent unauthorized access or disclosure.

d) Academic and Research Use Only

- The collected data will be used exclusively for academic research, project development, model training, testing, and validation.
- The data will not be sold, distributed, or used for commercial purposes.

e) Limited Access

- Access to the collected data will be restricted to the project team and faculty supervisor directly associated with the project.
- No data will be shared with third parties without prior approval from the concerned authorities.

4. Declaration

We hereby assure the clinic that all data collected as part of this project will be handled responsibly and ethically. Patient privacy and confidentiality will be maintained at all times, and no personally identifiable information will be disclosed in any report, presentation, publication, or demonstration arising from this project.

The data collected will be utilized solely for academic research and the development of the Fitonexa physiotherapy assistance system.

Submitted by:

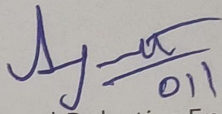
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